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Determination of the Resistance of Airways

pressure loss in a mine
Resistance of airways : airway.
Friction loss.



We have already discussed that for most of the mine airways, the pressure loss due to friction is

$$\Delta P_f = \frac{K \cdot S \cdot Q^2}{A^3} \quad \text{--- (1)}$$

For a given airway or a ventilation system such as a mine at any particular stage of working K , S and A can be taken as constant and hence the above equation can be written as

$$\Delta P_f = R_f Q^2 \quad \text{--- (2)}$$

Where R_f is called the resistance of the airway or the system

Combining equations (1) and (2) we have

$$R_f = \frac{K S}{A^3} = \frac{K \cdot P \cdot L}{A^3}$$

where K = coefficient of friction ($Ns^2/m^4 = Kg/m^3$)

S = Area of rubbing surface (m^2) = $L \cdot P$

A = cross sectional area of the roadway (m^2)

In addition to the pressure loss in airway due to frictional resistance offered by the airway, there is pressure loss due to change in cross-sectional area and presence of bends in the airway.

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Therefore, when determining the pressure loss in a projected mine airway, both friction and shock losses are to be taken into account.



Normally they are determined independently of each other, i.e. the friction loss for the airway without the obstruction is determined first and then shock loss due to the obstruction is determined and both are added together. Therefore,

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$$\text{Total pressure loss} = \overset{\Delta P_f}{\text{friction loss}} + \overset{\Delta P_s}{\text{shock loss}}$$

Slight bends or small area changes involving shock losses of only small magnitude often do not warrant their separate determination. They are usually included in the friction pressure loss by suitably increasing the value of the coefficient of friction K . Although both losses are mainly different from each other in the sense that friction losses are mainly due to surface drag and shock losses, to form drag.

Shock loss:

Shock loss is proportional to the velocity pressure corresponding to the average velocity of flow and is given by the relation

$$\Delta P_s = K \cdot P_v = K \cdot \frac{\rho \cdot v^2}{2} \quad \text{--- (3)}$$

where,

K = a dimensionless constant, called the shock factor.

P_v = Velocity pressure

Taking the standard value of $\rho = 1.2 \text{ kg/m}^3$, we have from above eqn.

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$$\Delta P_s = 0.6 \times v^2 = 0.6 \times \frac{Q^2}{A^2} \quad \text{--- (4)}$$

where v in m/s and Q in m^3/s

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Since for a given airway, the shock factor X and cross sectional area A are constant

$$\Delta P_s = R_s Q^2 \quad \text{--- (5)}$$



where $R_s = \text{shock resistance}$
 $= 0.6X/A^2$

In general terms, press. loss in a mine airway due to 'frictional work' ΔP can be given by the relation

$$\Delta P = \Delta P_f + \Delta P_s = \frac{KS V^2}{A} + 0.6X V^2 \quad \text{--- (6)}$$

Estimation of ΔP therefore requires a knowledge of K and X for any airway.

RESISTANCE OF AIRWAYS :

For all practical purposes, the total resistance of the airway

$$R = R_f + R_s \quad \text{--- (7)}$$

$$\text{or } R = \frac{KS}{A^3} + \frac{0.6X}{A^2} \quad \text{--- (8)}$$

The unit of R in the S.I. system is Ns^2m^{-8} .

The equivalence of some other prevalent units are :

$$1 \text{ Weisbach (Wb)} = 9.81 \text{ Ns}^2\text{m}^{-8}$$

$$1 \text{ Murg} = 0.00981 \text{ Ns}^2\text{m}^{-8}$$

$$1 \text{ Atkinson} = 0.00006 \text{ Ns}^2\text{m}^{-8}$$

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Determination of the Resistance of Airways

Resistance of airways can be determined from a knowledge of K and X with the help of the eq. (8).

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It is however better to determine the resistance of typical airways from field measurements of pressure drop and quantity of flow.

$$R = \frac{AP}{Q^2}$$

Resistance of a group of airways:

Resistance of a group of airways in simple series or parallel connection can be easily calculated, if resistance of each individual airway is known.

Usually there are three basic combinations of mine airways:

1. Airways in series
2. Airways in parallel and
3. Compound airways.

Combined resistance of airways in series:

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When the airways are connected in series, the ^{total} pressure drop across all of them is the sum of pressure drops across each of them, while the same quantity (vol. flow rate), Q flows through each of them.

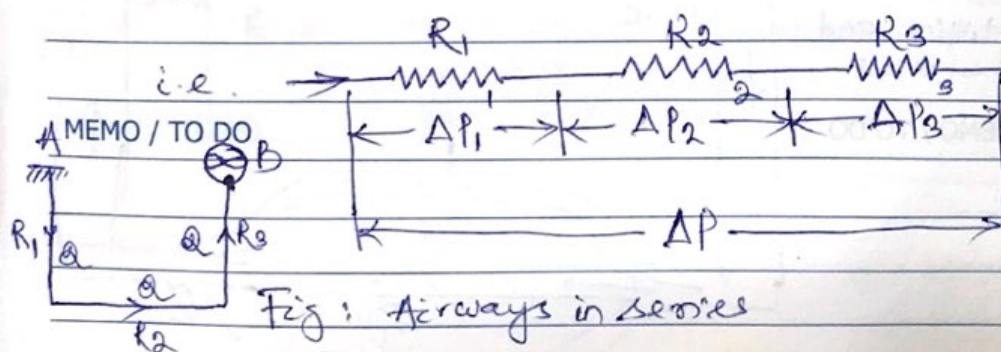


Fig: Airways in series

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Mine with 8 airways arranged

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i.e. $\Delta P = \Delta P_1 + \Delta P_2 + \Delta P_3$

Where $\Delta P = \text{Total pressure drop across the airways in series}$

$\Delta P_1 = \text{pressure drop in airway 1}$

$\Delta P_2 = \text{" " " 2}$

$\Delta P_3 = \text{" " " 3}$



If $R = \text{Combined / equivalent resistance of the airways in series}$
 $R_1, R_2 \text{ \& } R_3 = \text{Resistances of individual airways 1, 2 \& 3.}$

$$RQ^2 = R_1Q^2 + R_2Q^2 + R_3Q^2 = (R_1 + R_2 + R_3)Q^2$$

$$\Rightarrow R = R_1 + R_2 + R_3$$

If there are 'n' branches connected in series, then

$$R = R_1 + R_2 + R_3 + \dots + R_n$$

$$\Rightarrow R = \sum_{i=1}^n R_i$$

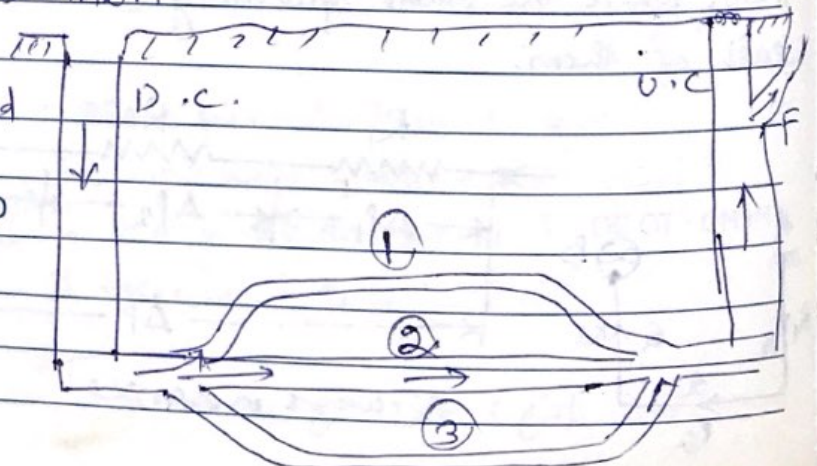
Combined resistance of airways in parallel

Airways are said to be connected in parallel when the total airflow is divided among them.
 Parallel connections between airways (also called splits) occur in a ventilation network where the airways branch at one point and rejoin at another.

* In mine ventilation, this practice is termed as splitting, and the branches are referred to as splits.

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MEMO / TO DO





In this case, the pressure drop across all of them is the same, say ΔP , while quantities get distributed amongst them depending on their resistances.

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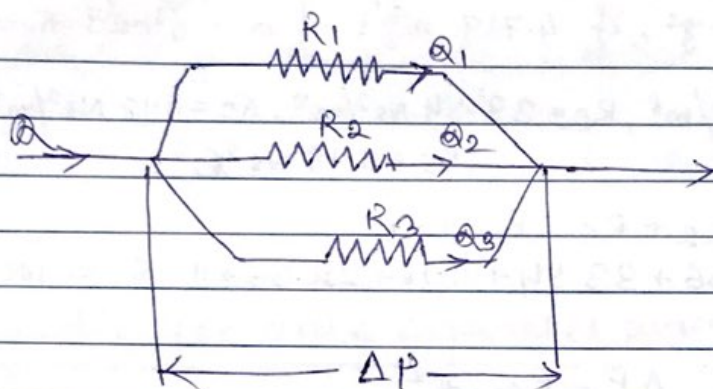


Fig: Airways in parallel.

$$\Delta P = R_1 \cdot Q_1^2 = R_2 \cdot Q_2^2, \dots, = R_n \cdot Q_n^2$$

Again Q , the total quantity of air flowing through all the airways is given by

$$\Rightarrow Q_1 = Q \sqrt{\frac{R}{R_1}}, Q_2 = Q \sqrt{\frac{R}{R_2}}, Q_3 = Q \sqrt{\frac{R}{R_3}} \leftarrow \text{Quantity divider rule in parallel airway}$$

$$Q = Q_1 + Q_2 + Q_3, \dots, Q_n$$

where $Q_1 = \sqrt{\frac{\Delta P}{R_1}}$, $Q_2 = \sqrt{\frac{\Delta P}{R_2}}$ and $Q_3 = \sqrt{\frac{\Delta P}{R_3}}$ etc.

$$\Rightarrow \sqrt{\frac{\Delta P}{R}} = \sqrt{\frac{\Delta P}{R_1}} + \sqrt{\frac{\Delta P}{R_2}} + \sqrt{\frac{\Delta P}{R_3}} + \dots + \sqrt{\frac{\Delta P}{R_n}}$$

$$\text{or, } \left[\frac{1}{\sqrt{R}} = \frac{1}{\sqrt{R_1}} + \frac{1}{\sqrt{R_2}} + \frac{1}{\sqrt{R_3}} + \dots + \frac{1}{\sqrt{R_n}} \right]$$

$$\text{or } R = \left(\frac{1}{\frac{1}{\sqrt{R_1}} + \frac{1}{\sqrt{R_2}} + \frac{1}{\sqrt{R_3}} + \dots + \frac{1}{\sqrt{R_n}}} \right)^2$$

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→ Discuss advantages of splitting of airways.

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Problem :
30 Given five airways in series with the following resistances (Ns^2/m^8). Find the equivalent resistance and the total head loss of the airways, if $4.719 \text{ m}^3/\text{s}$ of air flows through them.



$$R_1 = 22.36 \text{ Ns}^2/\text{m}^8, R_2 = 33.54 \text{ Ns}^2/\text{m}^8, R_3 = 11.18 \text{ Ns}^2/\text{m}^8, R_4 = 22.36 \text{ Ns}^2/\text{m}^8, R_5 = 11.18 \text{ Ns}^2/\text{m}^8.$$

Solⁿ.

$$R_{eq} = R_1 + R_2 + R_3 + R_4 + R_5 \\ = 22.36 + 33.54 + 11.18 + 22.36 + 11.18 = 100.6 \text{ Ns}^2/\text{m}^8$$

$$\text{Total head loss, } \Delta P = R_{eq} \cdot Q^2 \\ = 100.6 \times (4.719)^2 = 2240 \text{ Pa.}$$

Problem :

Four airways are arranged in parallel with a total quantity of $47.19 \text{ m}^3/\text{s}$ air flowing through them. The resistances of the airways are as follows:

$$R_1 = 2.627 \text{ Ns}^2/\text{m}^8, R_2 = 0.151 \text{ Ns}^2/\text{m}^8, R_3 = 0.349 \text{ Ns}^2/\text{m}^8 \text{ and } R_4 = 0.397 \text{ Ns}^2/\text{m}^8.$$

Find the head loss for the parallel airways and the quantity of air flowing through each.

Solⁿ.

$$R_{eq} = \left(\frac{1}{\sqrt{2.627}} + \frac{1}{\sqrt{0.151}} + \frac{1}{\sqrt{0.349}} + \frac{1}{\sqrt{0.397}} \right)^2 \\ = 0.0239 \text{ Ns}^2/\text{m}^8$$

$$\text{Total head loss, } \Delta P = R_{eq} \cdot Q^2 = 0.0239 \times (47.19)^2 \\ = 53.25 \text{ Pa.}$$

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$$\Delta P = R_1 \cdot Q_1^2 = R_2 \cdot Q_2^2 = R_3 \cdot Q_3^2 = R_4 \cdot Q_4^2$$

$$\Rightarrow Q_1 = \sqrt{\frac{\Delta P}{R_1}} = \sqrt{\frac{53.25}{2.627}} = 4.502 \text{ m}^3/\text{s}$$



$$Q_2 = \sqrt{\frac{\Delta P}{R_2}} = \sqrt{\frac{53.25}{0.151}} = 18.788 \text{ m}^3/\text{s}$$

$$Q_3 = \sqrt{\frac{\Delta P}{R_3}} = \sqrt{\frac{53.25}{0.349}} = 12.36 \text{ m}^3/\text{s}$$

$$Q_4 = \sqrt{\frac{\Delta P}{R_4}} = \sqrt{\frac{53.25}{0.397}} = 11.586 \text{ m}^3/\text{s}$$

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Cross check : Q should be equal to $Q_1 + Q_2 + Q_3 + Q_4$.

Quantity divider rule in parallel airways :

The quantity of air flowing through each parallel airway can be determined by knowing the total quantity and the resistance of each airway. Since the head losses for parallel airways are equal, the following can be written :

$$RQ^2 = R_1Q_1^2 = R_2Q_2^2 = R_3Q_3^2 = \dots$$

From this one can express an individual quantity as a function of the total circuit quantity (Q), the equivalent resistance (R) and the individual airway resistances. Thus

$$Q_1 = Q \sqrt{\frac{R}{R_1}}, \quad Q_2 = Q \sqrt{\frac{R}{R_2}} \quad \text{etc.}$$

It is apparent from the above equation that the airway with the highest value of resistance will have the smallest quantity of air flowing through it.

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Note : For some more problems refer page of 15, December, 2008 onwards

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Calculation of Shock Losses



When an airway changes its direction, due to bends, cross-sectional area or has an obstruction in ~~flow~~ like supports, there will be a loss of pressure due to disturbance in flow. This is called shock loss and normally expressed by which is proportional to the velocity pressure corresponding to the average velocity of flow and normally expressed by the relation

$$\Delta P_s = X \cdot P_v = X \cdot \frac{\rho \cdot v^2}{2}$$

Where ΔP_s = Shock loss (Pa)

P_v = Velocity pressure

X = Experimentally determined loss shock factor

ρ = air density (kg/m^3)

v = airflow velocity (m/s)

Shock factor

Shock factors for different shapes of obstructions, bends, area changes etc. are determined empirically mostly from model studies.

However, shock factors for any unique airway geometry occurring in the mine can be determined from field measurements with the help of above equation.

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MEMO / TO DO The shock factors can also be calculated as follows.

We know

$$\Delta P = \Delta P_f + \Delta P_s$$



$$\text{or } \Delta P_s = \Delta P - \Delta P_f$$

$$\text{But } \Delta P = P_1 - P_2 + g(h_1 - h_2) + \frac{v_1^2 - v_2^2}{2}$$

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$$\therefore \Delta P_s = P_1 - P_2 + g(h_1 - h_2) + \frac{v_1^2 - v_2^2}{2} - \Delta P_f$$

where P_1 and P_2 are absolute static pressures and can be measured by aneroid barometers.

But for better accuracy of measurement, a suitable water gauge connected by tubes to the two ends of the airway is used for measuring the pressure difference between the two ends. The tubes are usually made to face at right angles to the air-current so that the pressure difference measured is the static pressure difference.

We know that

Absolute static pressure = local atmospheric press. + gauge static press.

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$$\text{or } P = P_A + P_s$$

where P_A = local atmospheric pressure and
 P_s = gauge static pressure

Therefore, $P_1 = P_{A1} + P_{s1}$ and $P_2 = P_{A2} + P_{s2}$

Thus the above equation can be modified as

MEMO / TO DO

$$\Delta P_s = (P_{A1} + P_{s1}) - (P_{A2} + P_{s2}) + g(h_1 - h_2) + \frac{v_1^2 - v_2^2}{2} - \Delta P_f$$

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$$\text{Since } P_{A2} = P_{A1} + \rho \cdot g (h_1 - h_2)$$

then the eqⁿ. can be further simplified as



$$\Delta P_s = P_{s1} - P_{s2} + \frac{\rho}{2} (V_1^2 - V_2^2) - \Delta P_f$$

$$\text{Again since } P_s + \frac{\rho V^2}{2} = P_s + P_v = P_T$$

i.e. gauge static pressure + velocity pressure = Total pressure

the above equation further reduces to

$$\Delta P_s = (P_{T1} - P_{T2}) - \Delta P_f$$

The total pressure difference ($P_{T1} - P_{T2}$) can be directly measured on the watergauge if the tubes at the two ends of the airway are made to face the air-current, but the result will not be accurate since, the velocity pressure varies widely over the airway cross-section. That is why it is more common to measure the static pressure difference by the gauge-tube method by making the tube ends at right angles to the air-current and then determine the velocity pressure difference from the measured average velocities at both the ends of the airway by the equation

$$\frac{\rho}{2} (V_1^2 - V_2^2)$$

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It is of course necessary to independently estimate the value of ΔP_f for the airway over the measuring section.



Shock factor for bends

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Shock losses at bends are mainly due to the constriction of the cross-sectional

area of the air-stream at the bends and its subsequent expansion farther beyond the bend.

Right-angled bends or elbows are most commonly used in mines. Normal bends have the outer corner rounded by a radial curve centered on the diagonal and the inner corner either rounded concentrically or square. With the square bends, the outer corner is square, but the inner corner may be square or may form a radial curve centered on the diagonal. The inner bevel bend is only a variation of the square bend.

The approximate shock factor for normal bends is given by the relation

where

$$X = \frac{0.25}{R^2 \sqrt{a}} (2i/\pi)^2$$

X = Shock factor

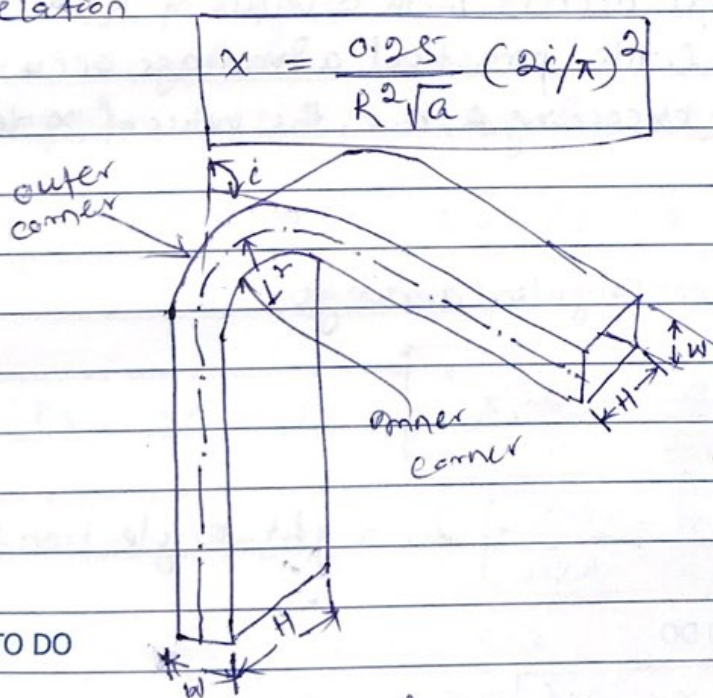
R = Radius ratio

a = aspect ratio

i = angle of deflection in rad.

$$\frac{R}{W} = R = \text{Radius ratio}$$

$$\frac{H}{W} = a = \text{aspect ratio}$$



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characteristics of a bend.

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For right-angle bends in rectangular airways,
the above equation becomes

$$X = \frac{0.25}{R^2 \sqrt{a}}$$



Since $i = \frac{\pi}{2}$ radian.

For square or circular airways, where $a = 1$

$$X = \frac{0.25}{R^2}$$

The value of R is minimum for a square inner corner and is equal to $\frac{W/2}{W} = 0.5$.

In such a case the value of X becomes the maximum ($= 1$).
 X decreases with increasing value of R and appreciable reduction in shock loss occurs with a value of $R = 0.15$ for which $X = 0.11$, but little practical advantage occurs from the value of R exceeding 2, i.e. the value of X decreasing below 0.06.

For square bends in rectangular airways,

$$X = \frac{0.6}{R \sqrt{a}} (2i/\pi)^2$$

which reduces to $X = \frac{0.6}{R \sqrt{a}}$ for right-angle bends and

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to

$$X = \frac{0.6}{R}$$

for right angle bends in
airways of square or circular cross-section.

Here also, the minimum value of R (for a square inner corner) is equal to 0.5 for which $x=1.2$.



This is the usual case with bends in mine airways.

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Example:

A mine airway, 2.1 m wide, 2.4 m high and 600 m long, contains a normal bend with a deflection of $\pi/3$ radian and a radius of curvature of 3 m. The airway passes a quantity of $9.5 \text{ m}^3/\text{s}$. Calculate the pressure loss in the airway as well as the resistance of the airway. Also, calculate the equivalent length of a straight airway of similar size and surface which will cause the same pressure loss. Assume the value of K to be equal to $0.0098 \text{ N s}^2/\text{m}^4$.

Solⁿ. Total pressure loss in the airway = Friction pressure loss + shock pressure loss
 or $\Delta P = \Delta P_f + \Delta P_s$

Friction pressure loss in the airway $\Delta P_f = \frac{K \cdot S \cdot Q^2}{A^3}$

or $\Delta P_f = \frac{0.0098 \times 2(2.1 + 2.4) \times 600 \times (9.5)^2}{(2.1 \times 2.4)^3} = 37.3 \text{ Pa}$

Shock pressure loss $\Delta P_s = 0.6 \cdot x \cdot \frac{Q^2}{A^2}$

Where x shock factor for bend = $\frac{0.25}{R^2 \sqrt{a}} (2i/\pi)^2$

MEMO / TO DO where $R = \text{radius ratio} = \frac{r}{w} = \frac{3}{2.1} = 1.43$

$a = \text{aspect ratio} = \frac{H}{W} = \frac{2.4}{2.1} = 1.143$

$i = \text{angle of deflection in rad} = \frac{\pi}{3}$

$\therefore x = \frac{0.25}{(1.43)^2 \sqrt{1.143}} \left(\frac{2 \cdot \frac{\pi}{3}}{\pi} \right)^2 = 0.0508$

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Therefore shock pressure loss due to the bend

$$\Delta P_s = 0.6 \times 0.0508 \left(\frac{9.5}{2.1 \times 2.4} \right)^2$$

$$= 0.108 \text{ Pa}$$



$$\therefore \text{Total pressure loss } \Delta P = 37.3 + 0.108 = 37.408 \text{ Pa}$$

$$\text{Since } \Delta P = R Q^2$$

$$\text{Then the resistance of the airway } R = \frac{\Delta P}{Q^2} = \frac{37.408}{(9.5)^2}$$

$$= 0.418 \text{ N s}^2 \text{ m}^{-8}$$

Equivalent length of straight airway (L)

$$\Delta P = \frac{K \cdot P \cdot L \cdot Q^2}{A^3} \Rightarrow L = \frac{\Delta P \cdot A^3}{K \cdot P \cdot Q^2} = \frac{37.408 \times (2.1 \times 2.4)^3}{0.0098 \times 2 \times (2.1 + 2.4) \times (9.5)^2}$$

$$= 601.7 \text{ m.}$$

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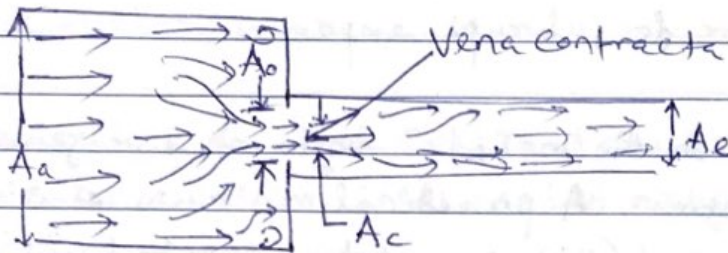
Shock factors for area change

Shock losses at area changes depend only on A_c (cross section of the flow-stream at the vena contracta) and A_e .

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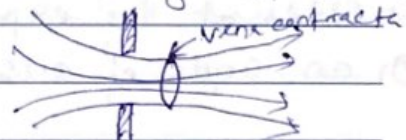
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Vena contracta : It is

the point in a fluid stream where the dia. of the stream is the least, and fluid velocity is at its maximum.



Since A_c depends on the values of A_a and A_0 , shock losses can be expressed by velocity pressures at any of these four areas.

Shock factor for abrupt symmetrical area changes :

$$X_0 = \left(\frac{1}{c} - N_e \right)^2$$

$$X_e = \left(\frac{1}{c} - N_e \right)^2 / N_e^2$$

$$X_a = \left(\frac{1}{c} - N_e \right)^2 / N_e^2$$

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Where X_0 , X_e and X_a are the shock factors corresponding to velocity pressures at sections with areas A_0 , A_e and A_a respectively.

MEMO / TO DO $C = \text{Coefficient of contraction} = \frac{A_c}{A_0}$

$N_e = \text{Ratio of area expansion} = \frac{A_0}{A_e}$

$N_c = \text{" of contraction} = \frac{A_0}{A_a}$

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